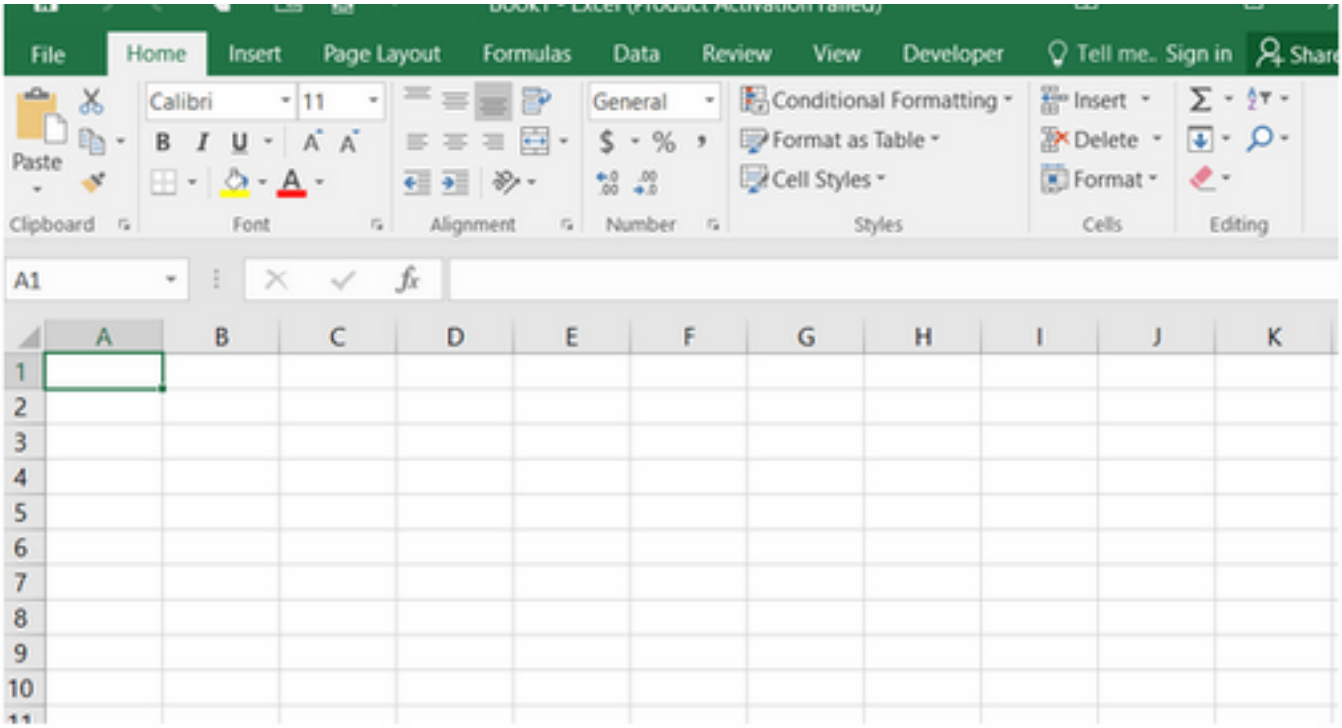
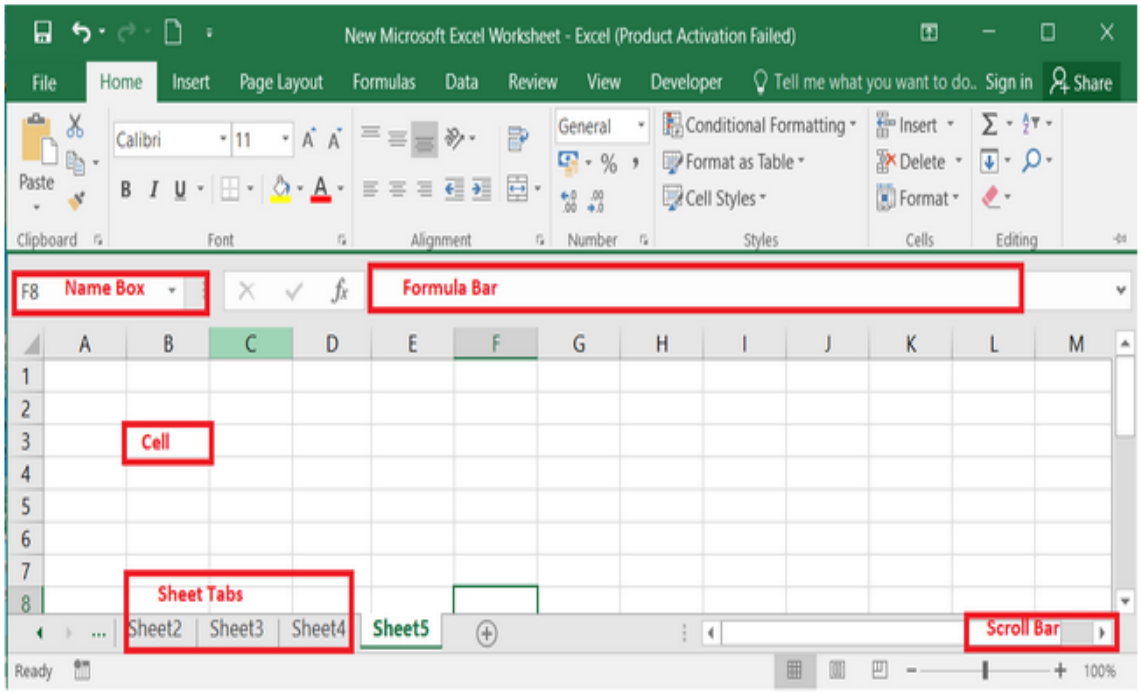
What is MS Excel

Microsoft Excel is a software application designed for creating tables to input and organize data. It provides a user-friendly way to analyze and work with data. The image below provides a visual representation of what an Excel spreadsheet typically appears like

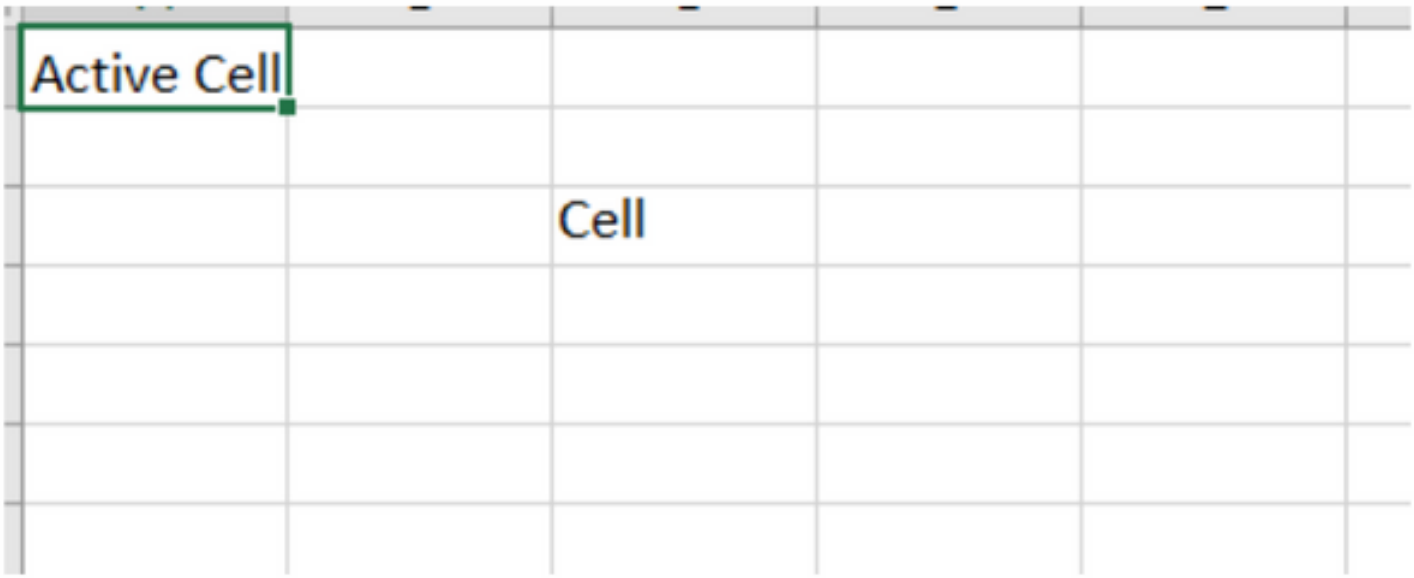


Excel Interface



What is a Cell

A spreadsheet takes the shape of a table, consisting of rows and columns. A cell is created at the intersection point where rows and columns meet, forming a rectangular box. Here's an image illustrating what a cell looks like:



What is Cell Address or Cell Reference

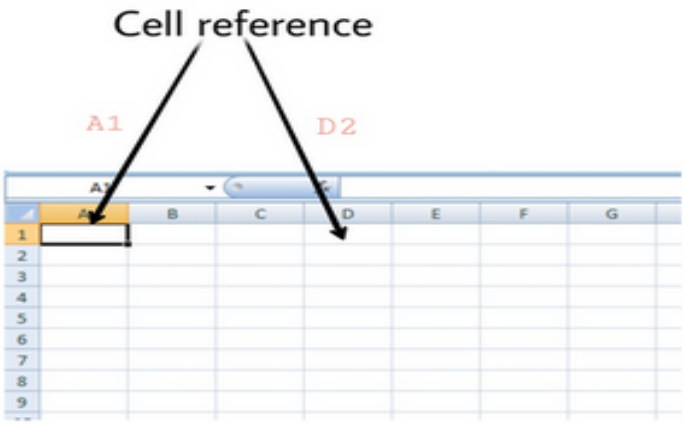
The address or name of a cell or a range of cells is known as Cell reference. It helps the software to identify the cell from where the data/value is to be used in the formula. We can reference the cell of other worksheets and

also of other programs.

- Referencing the cell of other worksheets is known as External referencing.
- Referencing the cell of other programs is known as Remote referencing.

There are three types of cell references in Excel:

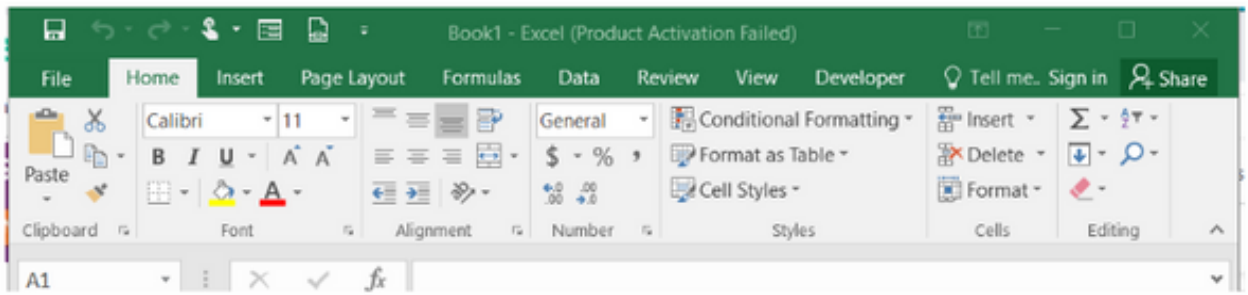
1. Relative reference.
2. Absolute reference.
3. Mixed reference.



Features of MS Excel

1. Ribbon

The Ribbon in MS-Excel is the topmost row of tabs that provide the user with different facilities/functionalities. These tabs are:



1. Home Tab

It provides the basic facilities like changing the font, size of text, editing the cells in the spreadsheet, autosum, etc.

1. Insert Tab

It provides the facilities like inserting tables, pivot tables, images, clip art, charts, links, etc.

1. **Page layout**

It provides all the facilities related to the spreadsheet-like margins, orientation, height, width, background etc. The worksheet appearance will be the same in the hard copy as well.

1. Formulas

It is a package of different in-built formulas/functions which can be used by user just by selecting the cell or range of cells for values.

1. Data

The Data Tab helps to perform different operations on a vast set of data like analysis through what-if analysis tools and many other data analysis tools, removing duplicate data, transpose the row and column, etc. It also helps to access data(s) from different sources as well, such as from Ms-Access, from web, etc.

1. Review

This tab provides the facility of thesaurus, checking spellings, translating the text, and helps to protect and share the worksheet and workbook.

1. View

It contains the commands to manage the view of the workbook, show/hide ruler, gridlines, etc, freezing panes, and adding macros.

Ms-Excel shortcuts

1. **Ctrl+N:** To open a new workbook.
2. **Ctrl+O:** To open a saved workbook.
3. **Ctrl+S:** To save a workbook.
4. **Ctrl+C:** To copy the selected cells.
5. **Ctrl+V:** To paste the copied cells.
6. **Ctrl+X:** To cut the selected cells.
7. **Ctrl+W:** To close the workbook.
8. **Delete:** To remove all the contents from the cell.
9. **Ctrl+P:** To print the workbook.
10. **Ctrl+Z:** To undo.

What is the Latest Versions of Excel

The most recent edition of Microsoft Excel is part of MS Office 2019, which is the latest version of the Microsoft Office suite. It comes packed with numerous new features introduced by Microsoft.

What is Ms Excel and its Features?

Microsoft Excel is a spreadsheet program that helps you manage and analyze data. Its key features include:

- Grid layout with cells for data.
- Formulas and functions for calculations.
- Charts and graphs for visualization.
- Data analysis tools like sorting and filtering.
- Collaboration and data protection options.

- Compatibility with various platforms.

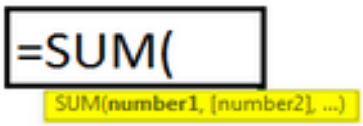
What is Ms Excel Definition?

Microsoft Excel is a computer program that lets you create tables to store and work with information, like numbers and words. It helps with tasks like calculations, making charts, and organizing data neatly.

List of Top 10 Basic Excel Formulas & Functions

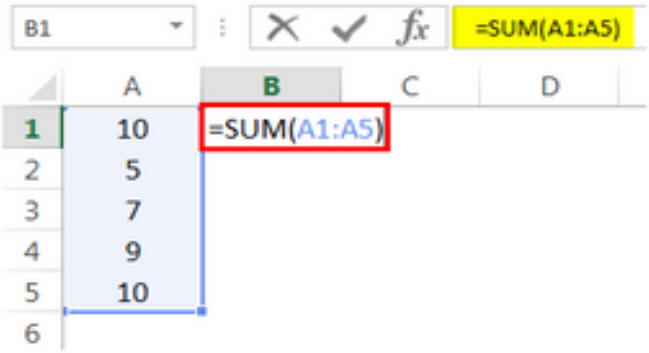
1. SUM
2. COUNT
3. COUNTA
4. COUNTBLANK
5. AVERAGE
6. MIN Excel
7. MAX Excel
8. LEN Excel
9. TRIM Excel
10. IF Excel

This basic Excel formula is used to get the sum of the value in one or more cells or ranges.

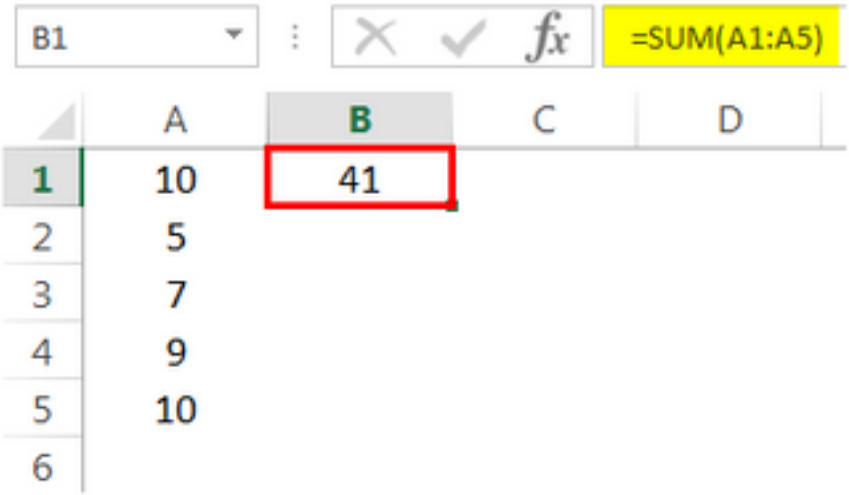


Example

=SUM(A1:A5**)**

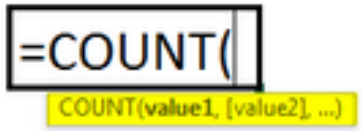


Result = 41 (See Image below)



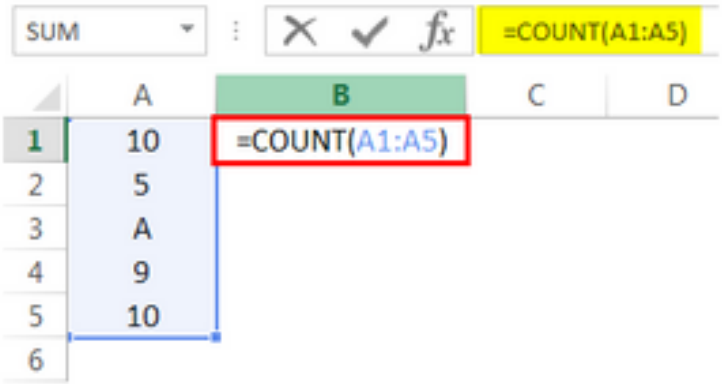
#2 COUNT Excel Function

This basic Excel function counts the numeric value in one or more cells or ranges.

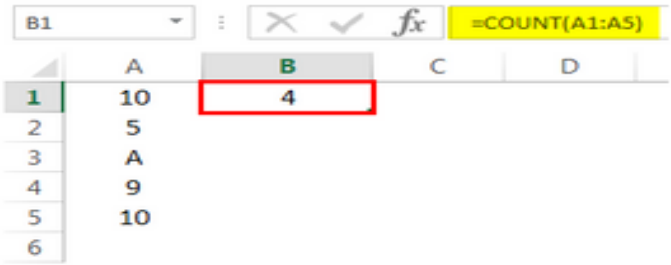


Example

=COUNT(A1:A5)

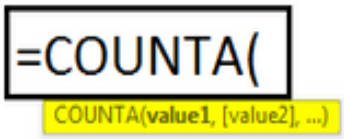


Result = 4 (This will exclude cell A3 since this formula calculates the only numeric value. Please see the image below).



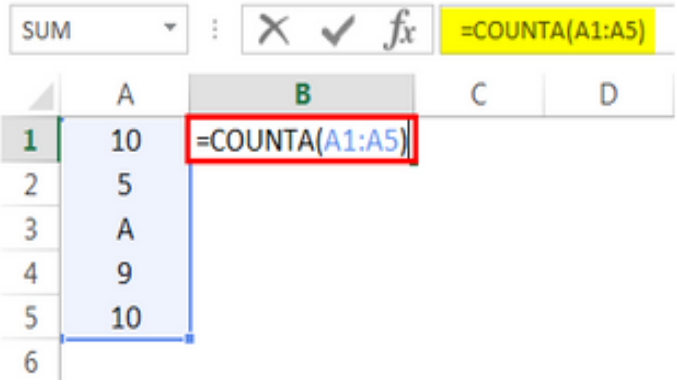
#3 COUNTA in Excel

This formula counts the value in one or more cells (This will measure the cells irrespective of the number or text value).

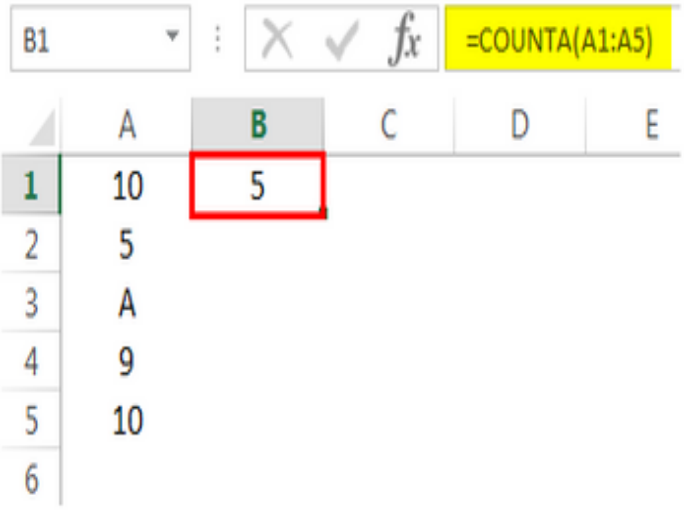


Example

=COUNTA(A1:A5)



Result = 5 (This will Include Cell A3 since this formula calculates both text and numeric value. See Image below).

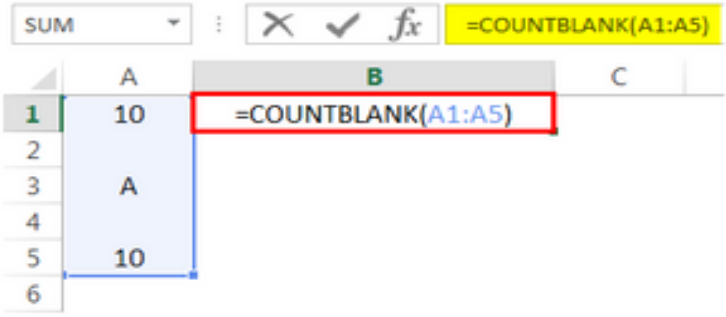


#4 COUNTBLANK in Excel

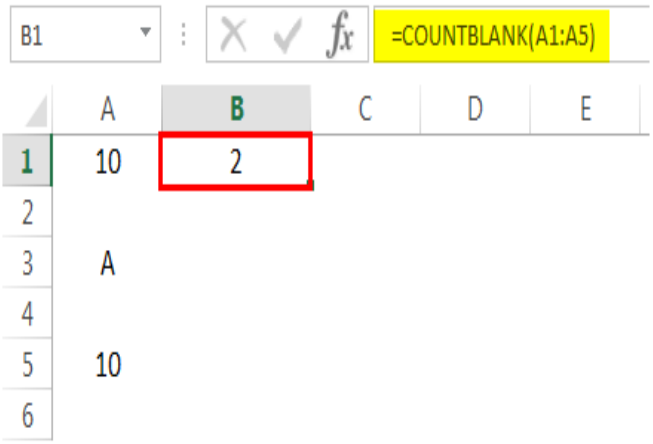
This Excel basic function counts the blank value in the range. (Note: We will not consider only space in a cell a blank cell).

Example

=COUNTBLANK(A1:A5)

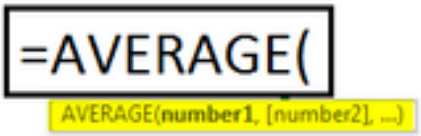


Result = 2 (This will calculate the number of blank cells in the range. See image below).



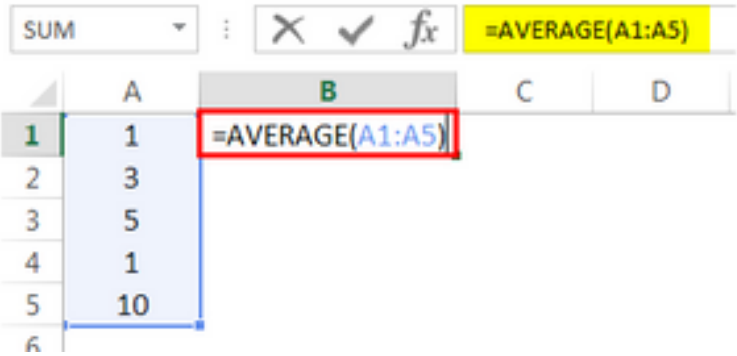
#5 AVERAGE in Excel

This basic formula in Excel is used to get the value average in one or more cells or ranges.

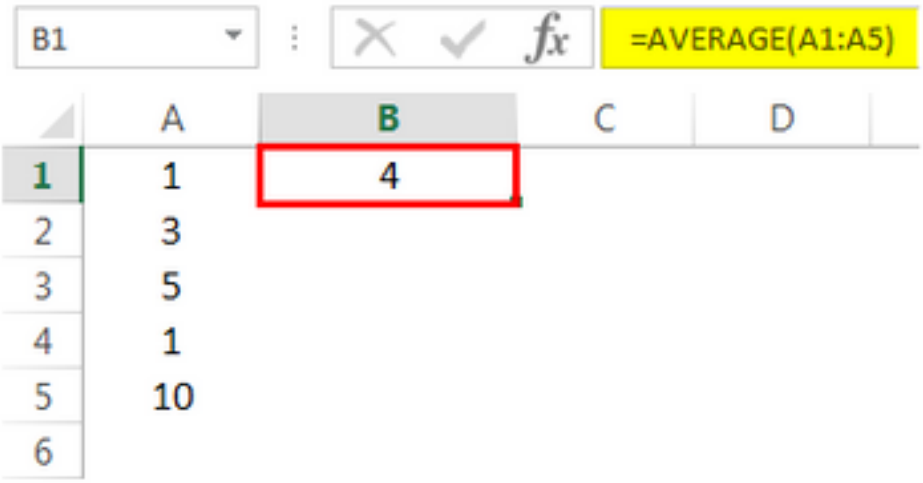


Example

=AVERAGE(A1:A5)

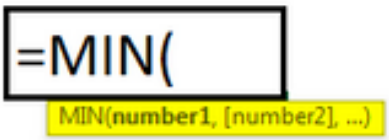


Result = 4 (See image below)



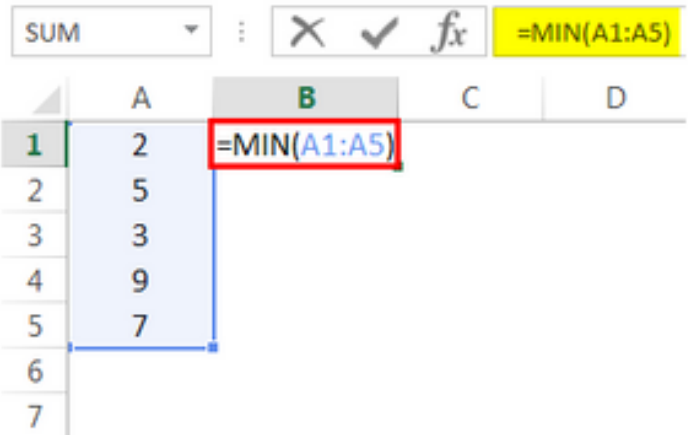
#6 MIN Formula in Excel

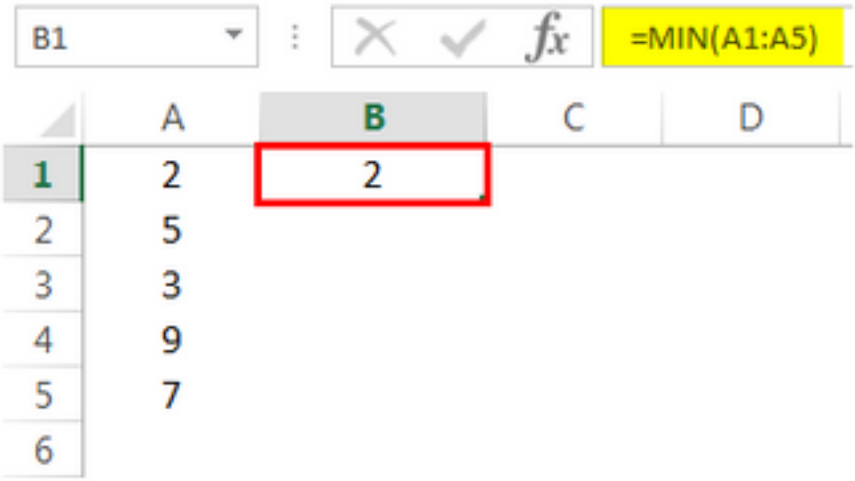
This Excel basic function is used to get the minimum value in cells or ranges.



Example

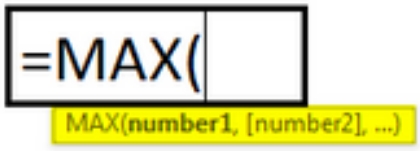
=MIN(A1:A5). Result = 2 (See Image below)





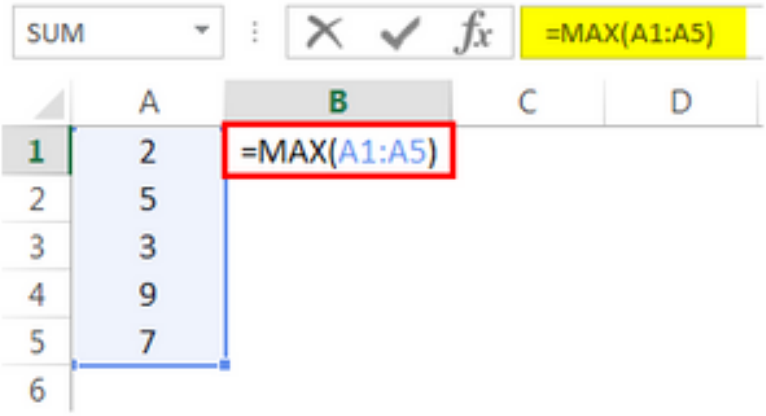
#7 MAX Formula in Excel

This basic Excel function gets the maximum value in cells or ranges.

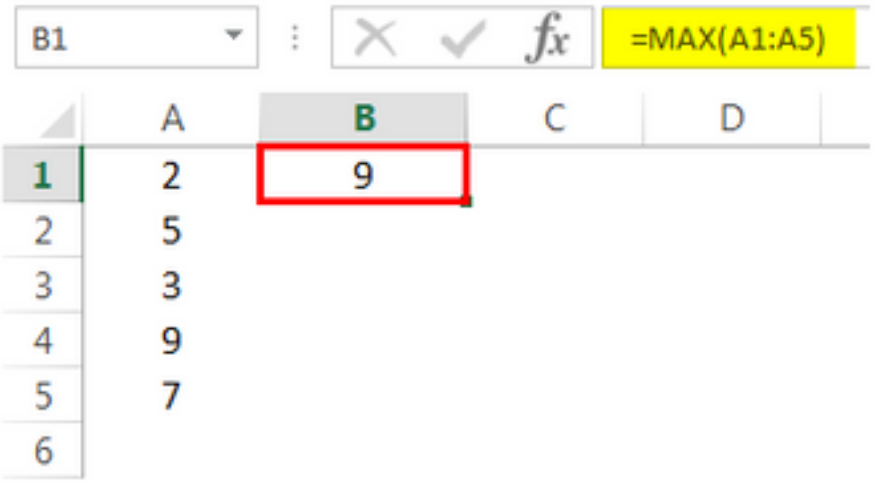


Example

=MAX(A1:A5)

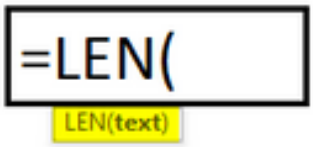


Result = 9 (See Image below)



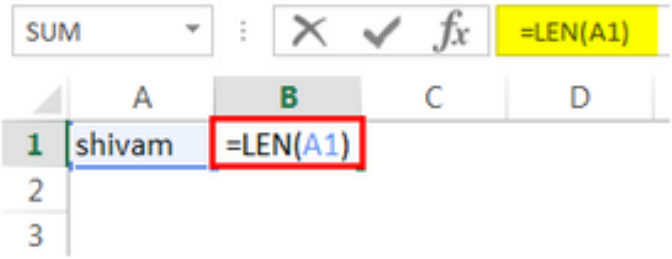
#8 LEN in Excel

This basic function of Excel calculates the number of characters in a cell or text.

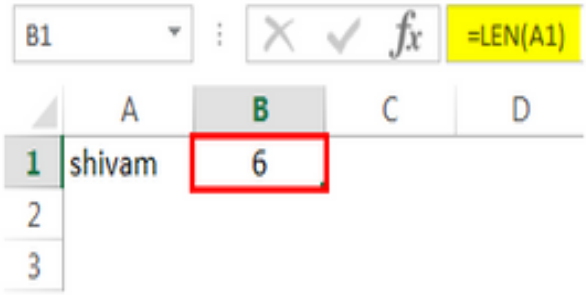


Example

=LEN(A1)



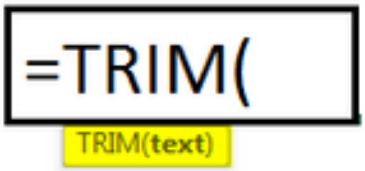
Cell A1 value is Shivam, which has a 6-character length. So, the result would be 6 (See image below).



#9 TRIM in Excel

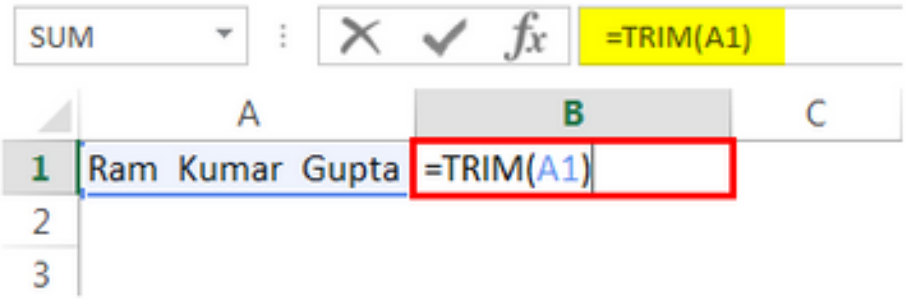
This basic Excel function removes unnecessary space in a cell or text.

SYNTAX: TRIM(TEXT)

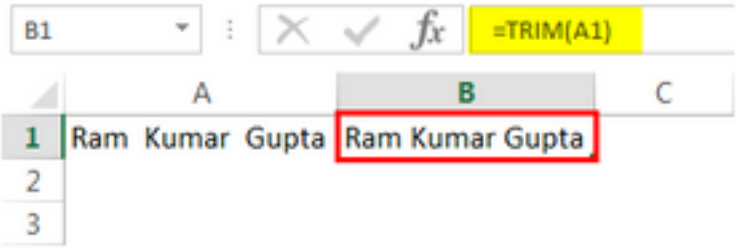


Example

=TRIM(A1)

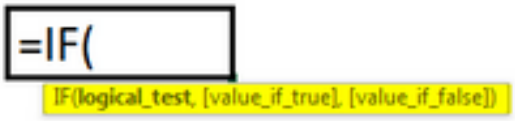


Cell A1 has two spaces between the first, middle, and last names. This function will remove extra space to give the result. (See image below).



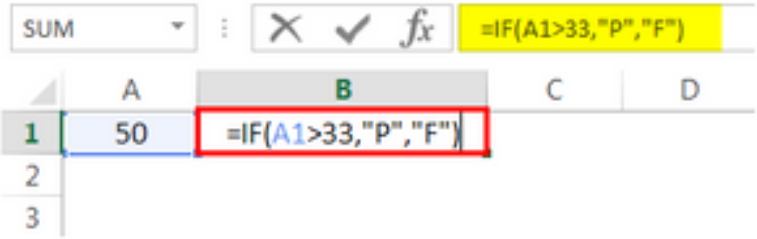
#10 IF in Excel

IF function can perform a logical test in Excel.

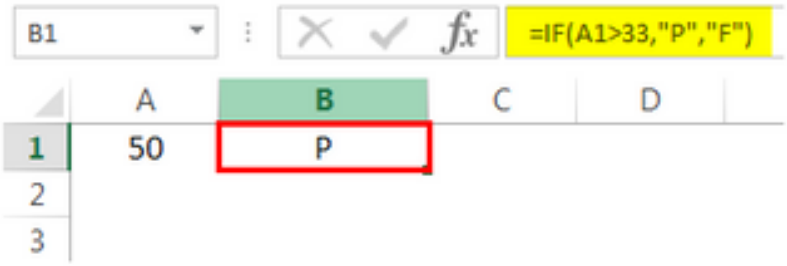


Example

=IF(A1>33,"P","F"). A value in cell A1 is 50, and a logical test is if the value is greater than 33. Then the result would be P. Else, the result would be F.



Since the value 50 is greater than 33, the result would be P. (See image below).



Excel Functions

Function	Description
=AND	Returns TRUE or FALSE based on two or more conditions
=AVERAGE	Calculates the average (arithmetic mean)
=AVERAGEIF	Calculates the average of a range based on a TRUE or FALSE condition
=AVERAGEIFS	Calculates the average of a range based on one or more TRUE/FALSE conditions
=CONCAT	Links together the content of multiple cells
=COUNT	Counts cells with numbers in a range
=COUNTA	Counts all cells in a range that has values, both numbers and letters
=COUNTBLANK	Counts blank cells in a range
=COUNTIF	Counts cells as specified
=COUNTIFS	Counts cells in a range based on one or more TRUE or FALSE condition
=IF	Returns values based on a TRUE or FALSE condition
=IFS	Returns values based on one or more TRUE or FALSE conditions
=LEFT	Returns values from the left side of a cell

=LOWER	Reformats content to lowercase
=MAX	Returns the highest value in a range
=MEDIAN	Returns the middle value in the data
=MIN	Returns the lowest value in a range
=MODE	Finds the number seen most times. The function always returns a single number
=NPV	The NPV function is used to calculate the Net Present Value (NPV)
=OR	Returns TRUE or FALSE based on two or more conditions
=RAND	Generates a random number
=RIGHT	Returns values from the right side of a cell
=STDEV.P	Calculates the Standard Deviation (Std) for the entire population
=STDEV.S	Calculates the Standard Deviation (Std) for a sample
=SUM	Adds together numbers in a range
=SUMIF	Calculates the sum of values in a range based on a TRUE or FALSE condition
=SUMIFS	Calculates the sum of a range based on one or more TRUE or FALSE condition

=SUMIFS	Calculates the sum of a range based on one or more TRUE or FALSE condition

=TRIM	Removes irregular spacing, leaving one space between each value
=VLOOKUP	Allows vertical searches for values in a table
=XOR	Returns TRUE or FALSE based on two or more conditions

Relative and Absolute Cell References

Introduction

There are two types of cell references: relative and absolute. Relative and absolute references behave differently when copied and filled to other cells. Relative references change when a formula is copied to another cell. Absolute references, on the other hand, remain constant no matter where they are copied.

Relative references

By default, all cell references are relative references. When copied across multiple cells, they change based on the relative position of rows and columns. For example, if you copy the formula =A1+B1 from row 1 to row 2, the formula will become =A2+B2. Relative references are especially convenient whenever you need to repeat the same calculation across multiple rows or columns.

To create and copy a formula using relative references:

In the following example, we want to create a formula that will multiply each item's price by the quantity. Rather than create a new formula for each row, we can create a single formula in cell D2 and then copy it to the other rows. We'll use relative references so the formula correctly calculates the total for each item.

1 Select the **cell** that will contain the formula. In our example, we'll select cell **D2.**

	Menu Item	Price	Quantity	Total
2	Empanadas: Beef Picadillo	$2.99	15	
3	Empanadas: Chipotle Shrimp	$3.99	10	
4	Empanadas: Black Bean & Plantain	$2.49	20	
5	Tamales: Chicken Tinga	$2.29	20	
6	Tamales: Vegetable	$2.29	30	
7	Arepas: Carnitas	$2.89	10	
8	Arepas: Queso Blanco	$2.49	20	
9	Empanadas: Apple Cinnamon	$3.19	40	
10	Beverages: Horchata	$1.89	25	
11	Beverages: Lemonade	$1.89	35	
12	Beverages: Tamarindo	$1.89	10	
13			Total	

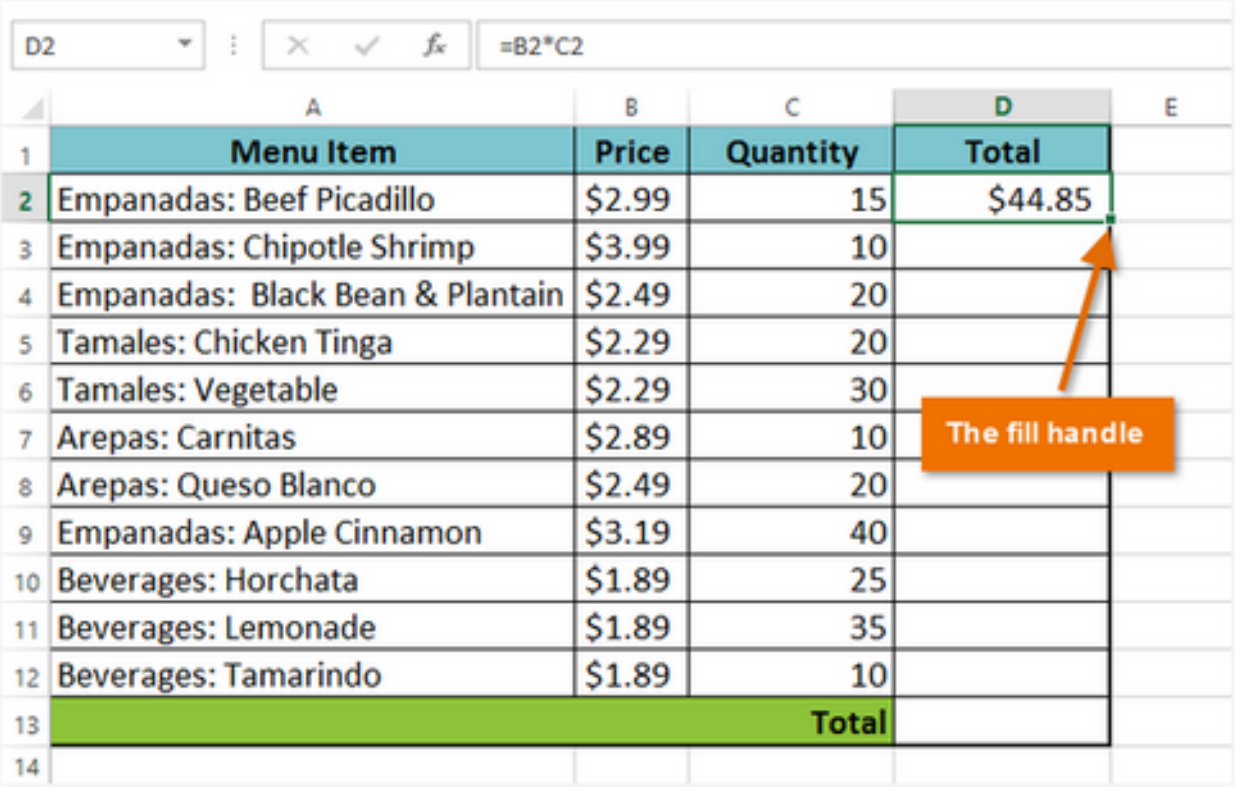
2 Enter the **formula** to calculate the desired value. In our example, we'll type **=B2*C2**.

C2 | =B2*C2

	A	B	C	D	E
1	**Menu Item**	**Price**	**Quantity**	**Total**	
2	Empanadas: Beef Picadillo	$2.99	15	=B2*C2	
3	Empanadas: Chipotle Shrimp	$3.99	10		
4	Empanadas: Black Bean & Plantain	$2.49	20		
5	Tamales: Chicken Tinga	$2.29	20		
6	Tamales: Vegetable	$2.29	30		
7	Arepas: Carnitas	$2.89	10		
8	Arepas: Queso Blanco	$2.49	20		
9	Empanadas: Apple Cinnamon	$3.19	40		
10	Beverages: Horchata	$1.89	25		
11	Beverages: Lemonade	$1.89	35		
12	Beverages: Tamarindo	$1.89	10		
13			**Total**		
14					

3 Press **Enter** on your keyboard. The formula will be **calculated**, and the result will be displayed in the cell.

4 Locate the **fill handle** in the lower-right corner of the desired cell. In our example, we'll locate the fill handle for cell **D2**.



	A	B	C	D	E
1	Menu Item	Price	Quantity	Total	
2	Empanadas: Beef Picadillo	$2.99	15	$44.85	
3	Empanadas: Chipotle Shrimp	$3.99	10		
4	Empanadas: Black Bean & Plantain	$2.49	20		
5	Tamales: Chicken Tinga	$2.29	20		
6	Tamales: Vegetable	$2.29	30		
7	Arepas: Carnitas	$2.89	10		
8	Arepas: Queso Blanco	$2.49	20		
9	Empanadas: Apple Cinnamon	$3.19	40		
10	Beverages: Horchata	$1.89	25		
11	Beverages: Lemonade	$1.89	35		
12	Beverages: Tamarindo	$1.89	10		
13			Total		
14					

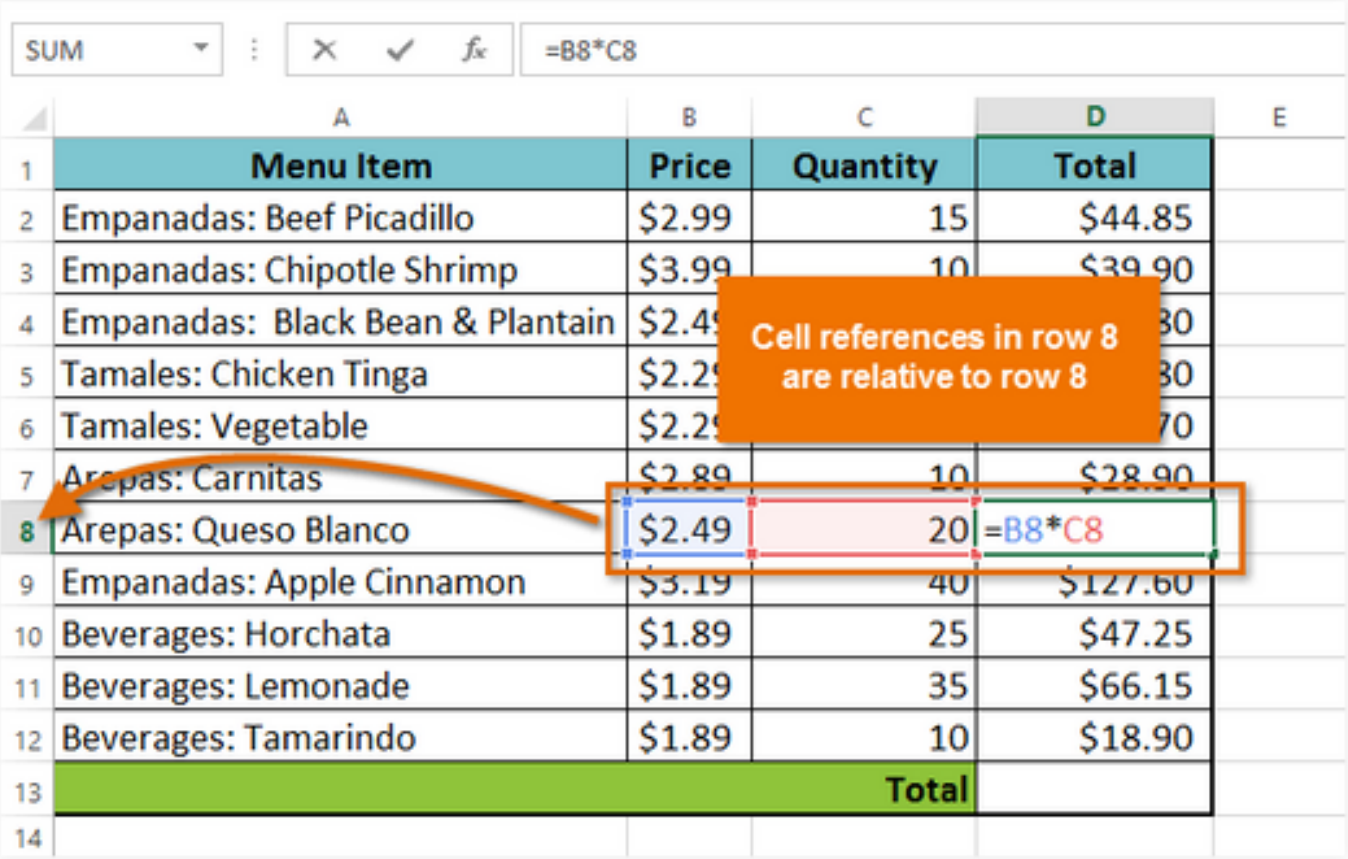
5 Click, hold, and drag the **fill handle** over the cells you wish to fill. In our example, we'll select cells **D3:D12**.

6 Release the mouse. The formula will be **copied** to the selected cells with **relative references** and the values will be calculated in each cell.

D2 | =B2*C2

	A	B	C	D	E
1	**Menu Item**	**Price**	**Quantity**	**Total**	
2	Empanadas: Beef Picadillo	$2.99	15	$44.85	
3	Empanadas: Chipotle Shrimp	$3.99	10	$39.90	
4	Empanadas: Black Bean & Plantain	$2.49	20	$49.80	
5	Tamales: Chicken Tinga	$2.29	20	$45.80	
6	Tamales: Vegetable	$2.29	30	$68.70	
7	Arepas: Carnitas	$2.89	10	$28.90	
8	Arepas: Queso Blanco	$2.49	20	$49.80	
9	Empanadas: Apple Cinnamon	$3.19	40	$127.60	
10	Beverages: Horchata	$1.89	25	$47.25	
11	Beverages: Lemonade	$1.89	35	$66.15	
12	Beverages: Tamarindo	$1.89	10	$18.90	
13			**Total**		
14					

You can double-click the **filled cells** to check their formulas for accuracy. The relative cell references should be different for each cell, depending on its row.



	Menu Item	Price	Quantity	Total
2	Empanadas: Beef Picadillo	$2.99	15	$44.85
3	Empanadas: Chipotle Shrimp	$3.99	10	$39.90
4	Empanadas: Black Bean & Plantain	$2.49	[illegible]	[illegible]30
5	Tamales: Chicken Tinga	$2.29	[illegible]	[illegible]30
6	Tamales: Vegetable	$2.29	[illegible]	[illegible]70
7	Arepas: Carnitas	$2.89	10	$28.90
8	Arepas: Queso Blanco	$2.49	20	=B8*C8
9	Empanadas: Apple Cinnamon	$3.19	40	$127.60
10	Beverages: Horchata	$1.89	25	$47.25
11	Beverages: Lemonade	$1.89	35	$66.15
12	Beverages: Tamarindo	$1.89	10	$18.90
13			Total	

Absolute references

There may be times when you do not want a cell reference to change when filling cells. Unlike relative references, absolute references do not change when copied or filled. You can use an absolute reference to keep a row and/or column constant.

An absolute reference is designated in a formula by the addition of a dollar sign ($) before the column and row. If it precedes the column or row (but not both), it's known as a mixed reference.

A2	The column and the row do not change when copied
A$2	The row does not change when copied
$A2	The column does not change when copied

- **You will use the relative (A2) and absolute (A2) formats in most formulas. Mixed references are used less frequently.**

To create and copy a formula using absolute references:

In our example, we'll use the 7.5% sales tax rate in cell E1 to calculate the sales tax for all items in column D. We'll need to use the absolute cell reference E1 in our formula. Because each formula is using the same tax rate, we want that reference to remain constant when the formula is copied and filled to other cells in column D.

1 Select the **cell** that will contain the formula. In our example, we'll select cell **D3.**

D3

	A	B	C	D	E
1			Sales Tax		7.5%
2	Menu Item	Price	Quantity	Sales Tax	Total
3	Empanadas: Beef Picadillo	$2.99	15		
4	Empanadas: Chipotle Shrimp	$3.99	10		
5	Empanadas: Black Bean & Plantain	$2.49	20		
6	Tamales: Chicken Tinga	$2.29	20		
7	Tamales: Vegetable	$2.29	30		
8	Arepas: Carnitas	$2.89	10		
9	Arepas: Queso Blanco	$2.49	20		
10	Empanadas: Apple Cinnamon	$3.19	40		
11	Beverages: Horchata	$1.89	25		
12	Beverages: Lemonade	$1.89	35		
13	Beverages: Tamarindo	$1.89	10		
14				Total	
15					

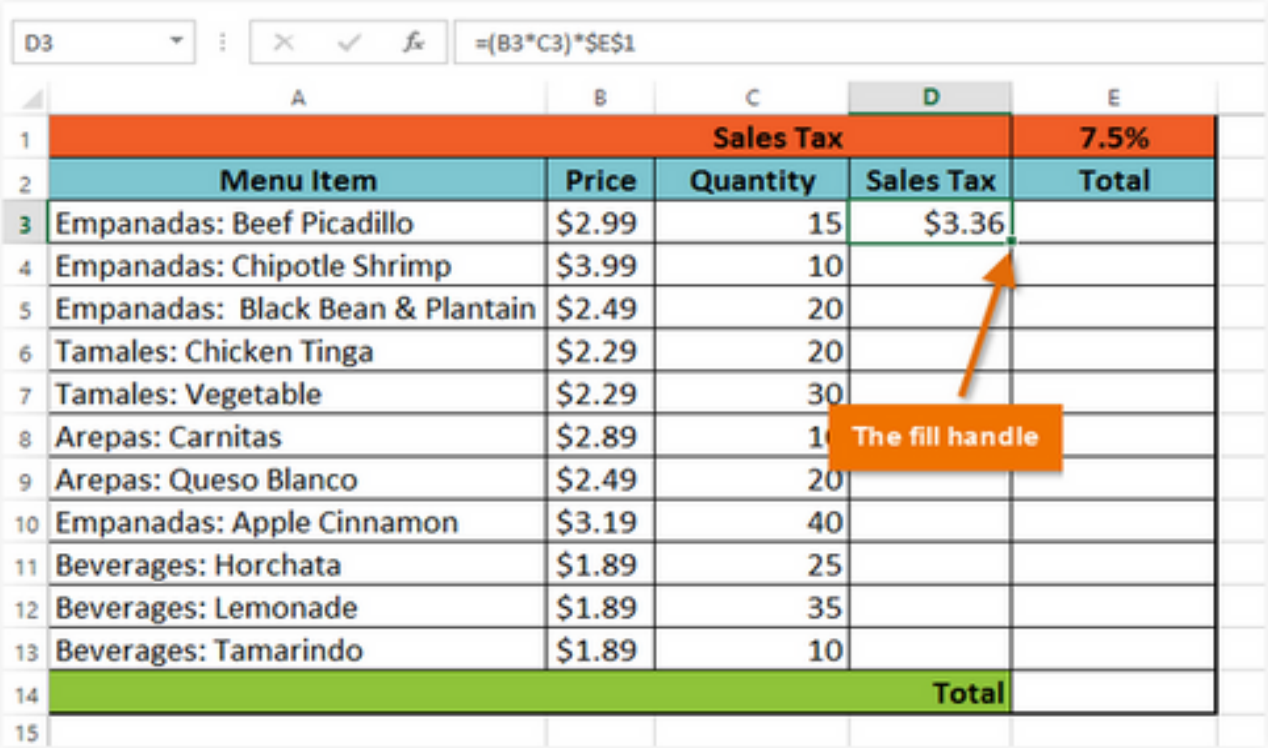
2 Enter the **formula** to calculate the desired value. In our example, we'll type **=(B3*C3)*E1**.

SUM | =(B3*C3)*E1

	A	B	C	D	E
1			Sales Tax		7.5%
2	Menu Item	Price	Quantity	Sales Tax	Total
3	Empanadas: Beef Picadillo	$2.99	=(B3*C3)*E1		
4	Empanadas: Chipotle Shrimp	$3.99	10		
5	Empanadas: Black Bean & Plantain	$2.49	20		
6	Tamales: Chicken Tinga	$2.29	20		
7	Tamales: Vegetable	$2.29	30		
8	Arepas: Carnitas	$2.89	10		
9	Arepas: Queso Blanco	$2.49	20		
10	Empanadas: Apple Cinnamon	$3.19	40		
11	Beverages: Horchata	$1.89	25		
12	Beverages: Lemonade	$1.89	35		
13	Beverages: Tamarindo	$1.89	10		
14				Total	
15					

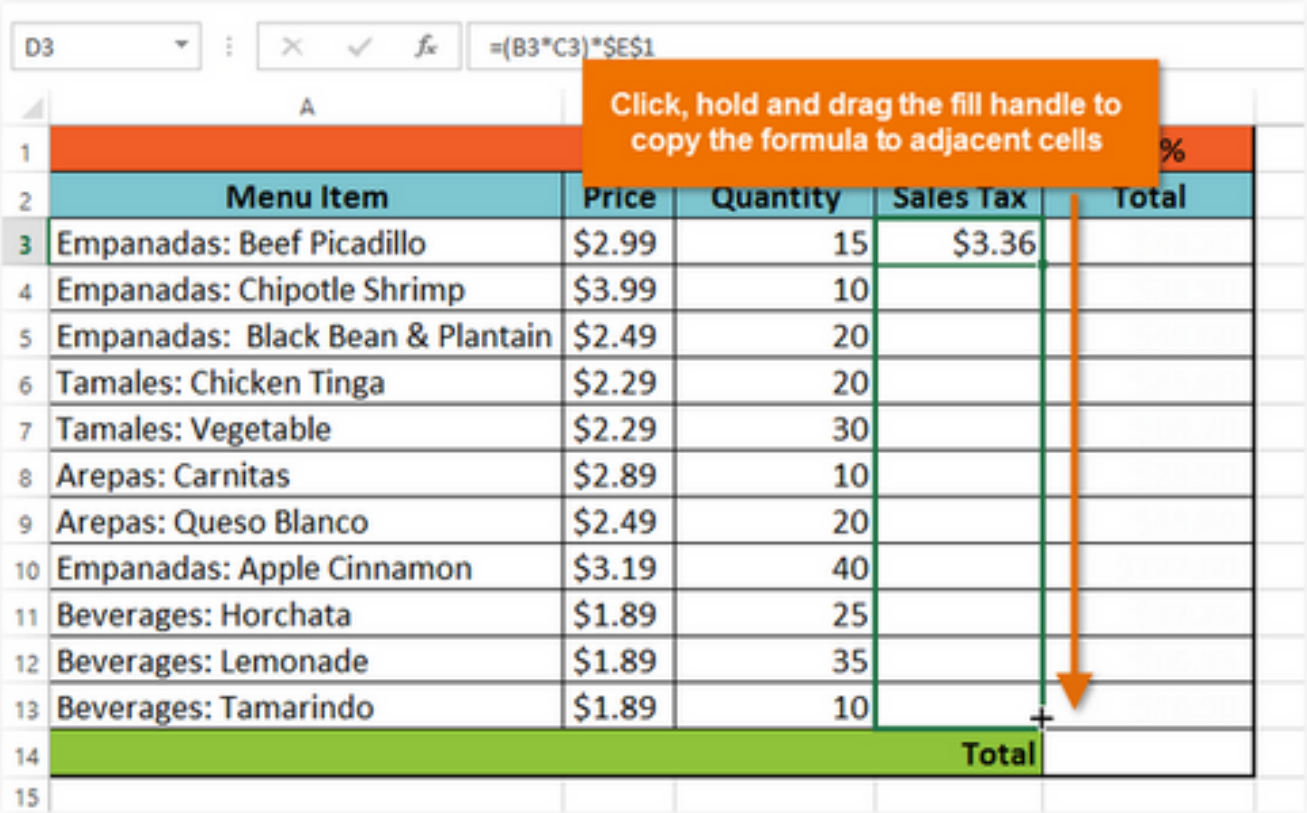
3. Press **Enter** on your keyboard. The formula will calculate, and the result will display in the cell.

4. Locate the **fill handle** in the lower-right corner of the desired cell. In our example, we'll locate the fill handle for cell **D3**.



	A	B	C	D	E
1			Sales Tax		7.5%
2	Menu Item	Price	Quantity	Sales Tax	Total
3	Empanadas: Beef Picadillo	$2.99	15	$3.36	
4	Empanadas: Chipotle Shrimp	$3.99	10		
5	Empanadas: Black Bean & Plantain	$2.49	20		
6	Tamales: Chicken Tinga	$2.29	20		
7	Tamales: Vegetable	$2.29	30		
8	Arepas: Carnitas	$2.89	1		
9	Arepas: Queso Blanco	$2.49	20		
10	Empanadas: Apple Cinnamon	$3.19	40		
11	Beverages: Horchata	$1.89	25		
12	Beverages: Lemonade	$1.89	35		
13	Beverages: Tamarindo	$1.89	10		
14				Total	
15					

5 Click, hold, and drag the **fill handle** over the cells you wish to fill, cells **D4:D13** in our example.

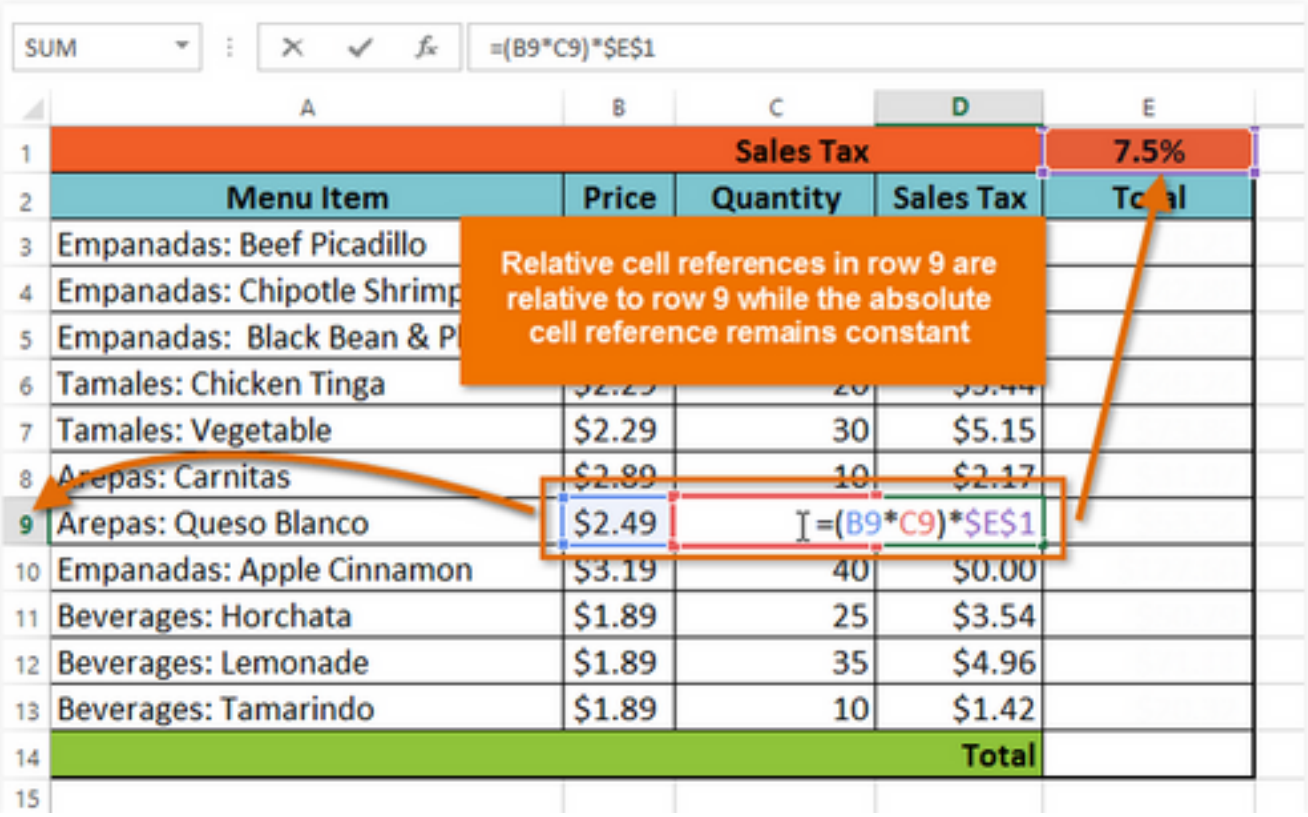


6 Release the mouse. The formula will be **copied** to the selected cells with an **absolute reference**, and the values will be calculated in each cell.

D3 =(B3*C3)*E1

	A	B	C	D	E
1			Sales Tax		7.5%
2	Menu Item	Price	Quantity	Sales Tax	Total
3	Empanadas: Beef Picadillo	$2.99	15	$3.36	
4	Empanadas: Chipotle Shrimp	$3.99	10	$2.99	
5	Empanadas: Black Bean & Plantain	$2.49	20	$3.74	
6	Tamales: Chicken Tinga	$2.29	20	$3.44	
7	Tamales: Vegetable	$2.29	30	$5.15	
8	Arepas: Carnitas	$2.89	10	$2.17	
9	Arepas: Queso Blanco	$2.49	20	$3.74	
10	Empanadas: Apple Cinnamon	$3.19	40	$9.57	
11	Beverages: Horchata	$1.89	25	$3.54	
12	Beverages: Lemonade	$1.89	35	$4.96	
13	Beverages: Tamarindo	$1.89	10	$1.42	
14				Total	
15					

You can double-click the **filled cells** to check their formulas for accuracy. The absolute reference should be the same for each cell, while the other references are relative to the cell's row.



Be sure to include the **dollar sign ($)** whenever you're making an absolute reference across multiple cells. The dollar signs were omitted in the example below. This caused the spreadsheet to interpret it as a relative reference, producing an incorrect result when copied to other cells.

SUM | =(B10*C10)*E8

	A	B	C	D	E
1			Sales Tax		7.5%
2	Menu Item	Price	Quantity	Sales Tax	Total
3	Empanadas: Beef Picadillo	$2.99	15	$3.36	$48.21
4	Empanadas: Chipotle Shrimp	$3.99	10	#VALUE!	$42.89
5	Empanada			$2,401.04	$53.54
6	Tamales: C			#VALUE!	$49.24
7	Tamales: V			$168,373.03	$73.85
8	Arepas: Ca			#VALUE!	$31.07
9	Arepas: Qu			$8,388,398.27	$53.54
10	Empanadas: Apple Cinnamon	$3.19	40	=(B10*C10)*E10	$127.60
11	Beverages: Horchata	$1.89	25	$396,354,176.00	$50.79
12	Beverages: Lemonade	$1.89	35	#VALUE!	$71.11
13	Beverages: Tamarindo	$1.89	10	$7,491,094,819.49	$20.32
14				Total	

Without the dollar sign ($), the reference to cell E1 was interpreted as a relative reference, leading to incorrect results

Data Manipulation Techniques

Data analysis is a challenging task, especially if you don't have the data manipulation skills. In this article, we will discuss some of the most common data manipulation techniques that can be used in Excel for data analysis.

The power of these techniques will be demonstrated by using some real-life examples. The 9 common data manipulations techniques discussed are:

1) Filtering

2) Sorting

6) Changing Data Time

7) Adding Columns and Rows

8) Naming Columns or Rows

9) Inserting Columns or Rows.

Filtering

Filtering is a process of sorting data by a certain criteria. It's an effective way to identify subsets of data from the larger dataset.

Lets say, we have a dataset with monthly sales data from 2020–2023 and lets say you want to see the sales number for only 2023. Excel filter function is useful if you want to see the total sales for the year only, or if you want to know how many months had positive growth.

First, identify which column contains the filter criteria that will be used to filter your data. In this case, we can create a new column called "Sales Growth." Next, highlight your "Sales Growth" column and select "Filter" from the Data menu on the toolbar. This will open a dialogue box where you can input your filtering criteria. In this example, we are using ">0%" as my filter criterion to calculate total sales for years with positive revenue growth > 0%.

Sorting

Sorting is another technique of data analysis and is used to rearrange the order of your data. It's an easy way of exploring and understanding your data.

For example, let's say you had a list of 5 different numbers:

1, 10, 2, 3, 4

If you want to know how to sort in excel, here's an example: You want to sort this list in ascending order (from lowest to highest), we would click on the column heading for this list and then select "Sort Ascending". This will arrange the list like this:

1, 2, 3, 4, 10.

By sorting the numbers in ascending order we can see that they are increasing in size. If we wanted to change the sort order to descending (highest to lowest) we would click on Column A and select "Sort Descending" like so:

10, 4, 3, 2, 1.

Again by sorting in descending order from highest number to lowest number we can see that they're decreasing in size.

S.NO.	HLOOKUP	VLOOKUP
01.	It assumes that data is arranged as a table with different elements of the information in different rows.	Whereas it assumes that data is arranged as a table with different elements of the information in different columns.
02.	It looks for a value horizontally across the lookup table.	It looks for a value vertically across the lookup table.
03.	The use of HLookup function is less as compared to VLookup function.	The use of VLookup function is more than compared to HLookup function.
04.	It is used when data is listed in rows.	Whereas it is used when data are listed in columns.
05.	In HLookup function, the value the user is searching in the table or range	In VLookup function, the value the user is searching must be located in the left-

05.	In HLookup function, the value the user is searching in the table or range must be in the top-most column.	In VLookup function, the value the user is searching must be located in the left-most column of the table or range.

06.	HLookup function returns the searched value in the same column.	VLookup function returns the searched value in the same row but in the next column.
07.	The H in the HLookup function stands for horizontal search	The V in VLookup function stands for Vertical search.
08.	It is used to find out data from a range in the bottom-most range.	It helps to find out the data in the leftmost column.